

MATHS

DISCOUNT

(छूट)

**ALL TYPE WISE
QUESTIONS**



Maths with Gajanand

3. DISCOUNT

mark Price Selling Price
 Rs. 500 Rs. 400



Discount = 100

$$* \quad \text{Discount \%} = \frac{\text{Discount}}{\text{M.p}} \times 100 \quad *$$

$$\text{D \%} = \frac{100}{500} \times 100$$

$$\text{D \%} = \boxed{20\%}$$

Q.1. If a dining table with marked price Rs. 6,000 was sold to a customer for Rs. 5,520, then the rate of discount allowed on the marked price of the table is:

Soln

M.p S.p
 6000 5,520



$$\text{D \%} = \frac{6000 - 5520}{6000}$$

$$= \frac{480}{6000} \times 100$$

$$= \boxed{8\%} \quad \underline{\underline{\text{Ans}}}$$

[2] After a discount of 34% an article is sold for Rs 3168
What is the marked price (in Rs) of the article?

Soln :- [m-I]

$$x \times \frac{(100-34)}{100} = 3168$$

$$x \times \frac{66}{100} = 3168$$

$$x = \boxed{4800 \text{ Rs}}$$

[m-II]

$$M.P = 100$$

$$M.p = \frac{3168}{66} \times 100$$

$$= \boxed{4800} \text{ Ans}$$

[3] A distributor gives a discount of 12% on the marked Price. If the selling price is Rs. 9240, then what will be the discount on it?

Soln :- $12\% = \frac{3}{25} \leftarrow \text{Discount}$
 $25 \leftarrow M.P$

$$\text{discount} = \frac{9240}{(25-3)} \times 3$$

$$= \frac{9240 \times 3}{22}$$

$$= \boxed{1260} \text{ Ans}$$

[4] A house was sold for ₹ y by giving a discount of x%, then the list price was ?

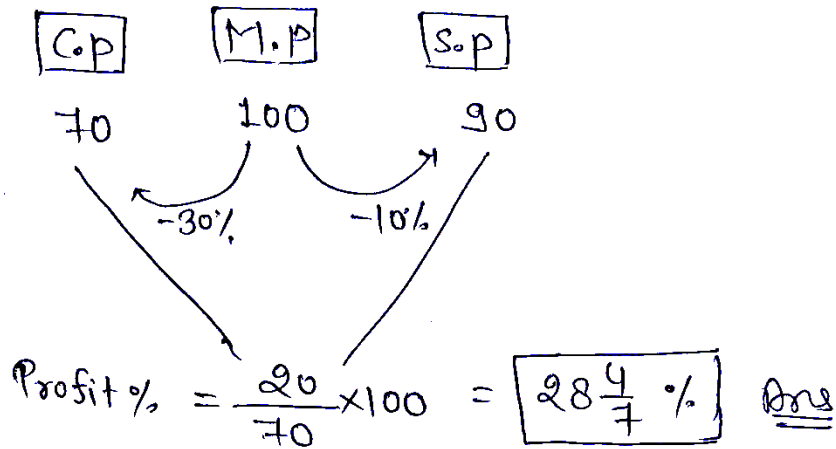
Soln :-

$$M.p \times \frac{(100-x)}{100} = y$$

$$\boxed{M.p = \frac{100y}{100-x}} \text{ Ans}$$

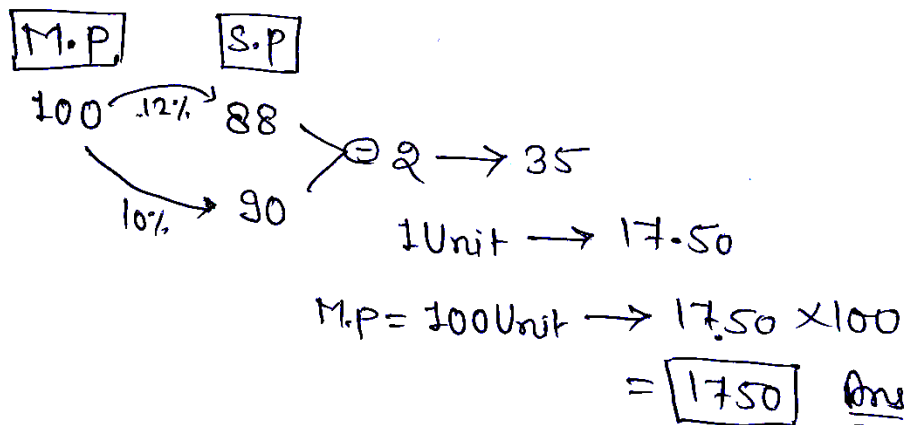
- [5] A book seller allowed 10% discount on Printed Price. He gets 30% commission from publisher. His Profit in percent will be ?

Soln

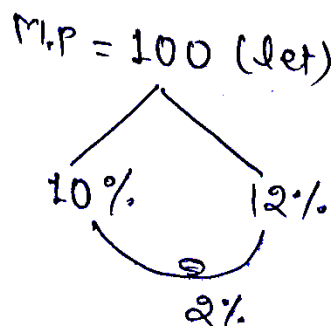


- [6] While selling a cooler, a shopkeeper gives a discount of 10% on the marked Price. If he gives a discount of 12% he earns ₹ 35 less as profit. The marked Price of the cooler is ?

Soln :-



m-II



$$\frac{35}{2\%} \times 100\%$$

$$M.P. = 1750 \quad \underline{\text{Ans}}$$

9] A dealer is selling an article at a discount of 5% on the marked price. If the marked price is 12% above the cost price and the article was sold for ₹ 532 then the cost price is ?

Soln:-

$C.P \xrightarrow{+12\%} M.P \xrightarrow{-5\% = -5.6} S.P$
 $100 \quad 112 \quad 106.4 \longrightarrow 532$
 $1 \text{ Unit} \longrightarrow 5$
 $C.P = 100 \text{ Unit} \longrightarrow \boxed{500} \text{ Ans}$

10 The selling price of an article is Rs. 816. If the discount on it is 15%. What would be the discount on it is 15%. What would be the selling price of the article if the discount on it is 25%?

Soln. - m-1

M.P S.P

$\frac{48}{816} \times 20$ 816

$\frac{85}{85} \times 100$

$= 48 \times 20$

$\swarrow$ -25% $\frac{(48 \times 20) \times 75}{100}$

$= \boxed{720}$ Ans

m-II

$$15\% = \frac{3}{20} \leftarrow \begin{array}{l} \text{Discount} \\ \text{M.P} \end{array}$$

$$S.P = 20 - 3 = 17 \text{ Unit} \rightarrow 816$$

$$M.p = 48 \times 20$$

$$\begin{aligned} \text{New S.P} &= 48 \times 20 \times \frac{3}{4} \quad \leftarrow \begin{array}{l} \text{Discount} \\ 25\% \end{array} \\ &= \boxed{720} \quad \underline{\text{Ans}} \end{aligned}$$

27-III Shortcut

$$\frac{816}{(100-15\%)} \times (100-25\%) \Rightarrow \frac{816}{85} \times 75\% \Rightarrow \boxed{720} \text{ Ans}$$

- 11** If 60% discount is offered on the marked Price and selling price becomes equal to cost price then what is the % mark up?

Soln:-

$$\begin{array}{ccc} \text{M.P} & & \text{S.P} = \text{C.P} \\ 100 & & 40 \end{array}$$

$$\frac{60}{40} \times 100 = \boxed{150\%} \quad \underline{\text{Ans}}$$

- 12** A dealer buys an article at a discount of 20% on its list Price and marks it at 25% above the list Price. If he allows a 20% discount on the new list Price, then his profit percent is?

Soln:-

$$\begin{array}{ccccccc} & & +25\% & & \text{Discount } 20\% & & \\ & & \nearrow & & \searrow & & \\ \text{C.P} & \text{M.P} & \text{New M.P} & & \text{S.P} & & \\ 80 & 100 & 125 & & 100 & & \end{array}$$

$$\text{Profit \%} = \frac{20}{80} \times 100 = \boxed{25\%} \quad \underline{\text{Ans}}$$

- 13** A shopkeeper sells an item for Rs. 492 after allowing 18% discount on its marked Price. Had he NOT allowed any discount, he would have earned a profit of 20% on the cost price of the item?

Soln:-

$$\begin{array}{ccc} \text{C.P} & \text{M.P} & \text{S.P} \\ & \frac{492}{82} \times 100 & 492 \\ \frac{600}{120} \times 100 & \nearrow & \\ = \boxed{500} & = 600 & \end{array}$$

m-II $18\% = \frac{9}{50}, 20\% = \frac{1}{5}$

$$\left[\frac{492}{41} \times 50 \right] \times \frac{5}{6}$$

M.P

$$\text{C.P} = \boxed{500} \quad \underline{\text{Ans}}$$

[14] If the selling price of an article is 32% more than its cost price and the discount offered on its marked price is 12%, then what is the ratio of its cost price to the marked price?

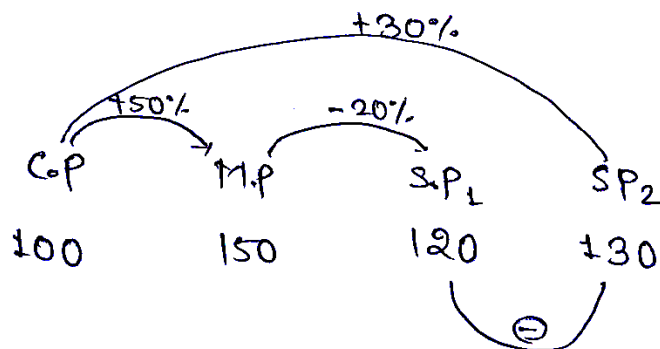
Soln:-

C.P	S.P	M.P	$32\% = \frac{8}{25}$
25×2	33×2		$12\% = \frac{3}{25}$
	22×3	25×3	
50	66	75	

C.P : M.P = 50 : 75 = 2 : 3 Ans

[15] Anand marks up the price of an article by 50% and then allows a discount of 20% and sells it to Balaji who sells it for ₹ 20 more than what he purchased for, this S.P is 30% more than the original C.P of the article. Then Balaji's profit % is?

Soln:-



10 Unit $\rightarrow$ ₹ 20

1 Unit $\rightarrow$ ₹ 2

C.P for balaji 120 Unit $\rightarrow 120 \times 2$
 $= 240$

% Profit = $\frac{\text{₹ } 20}{240} \times 100$

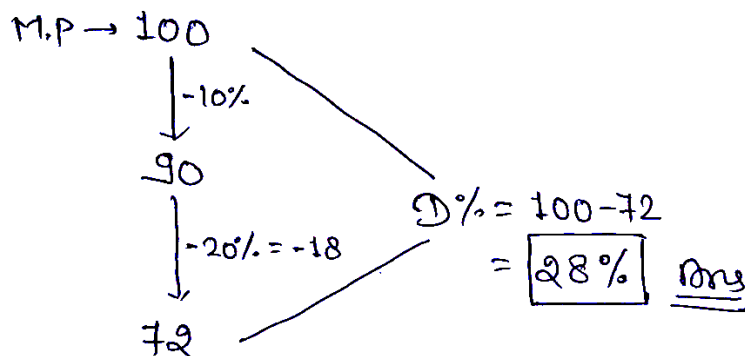
= 8.33% Ans

Successive Discounts

Different ways to find Effective/Equivalent Discount

Ex. 10% , 20%

m-I Basic



m-II Formula

$$x + y - \frac{xy}{100}$$

$$10 + 20 - \frac{10 \times 20}{100}$$

$$= \boxed{28\%}$$
 Ans

m-III $10\% = \frac{1}{10}$, $20\% = \frac{1}{5}$

Fraction method

M.P	S.P
10	9
$\frac{5}{50}$	$\frac{4}{36}$

$$\frac{14}{50} \times 100 = \boxed{28\%}$$
 Ans

m-IV

$$M.P \times \frac{9}{10} \times \frac{4}{5} = S.P$$

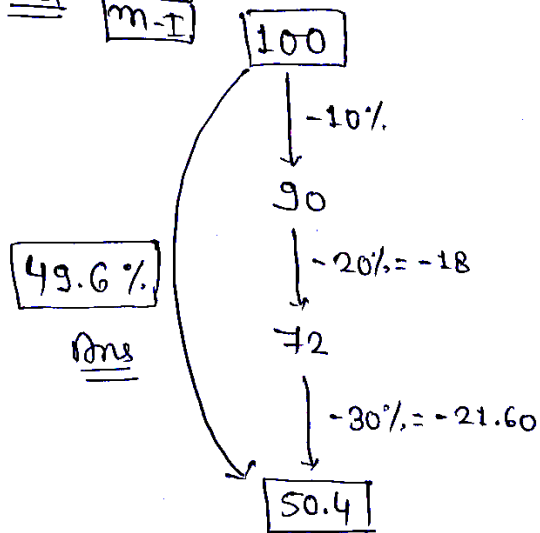
$$\frac{M.P}{S.P} = \frac{25}{18} \Rightarrow 7$$

$$\text{Discount \%} = \frac{7}{25} \times 100 = \boxed{28\%}$$
 Ans

Ex. 10% , 20% , 30% overall Discount ?

Solⁿ

m-I



m-II

Fraction method

M.P	S.P
10	9
5	4
10	7
500	252

$\frac{248}{500} \times 100 = 49.6\%$
Ans

m-III

Formula

$$x + y + z - \frac{(xy + yz + zx)}{100} + \frac{xyz}{10000} \%$$

$$= 10 + 20 + 30 - \frac{(200 + 600 + 300)}{100} + \frac{6000}{10000}$$

$$= 60 - 11 + 0.6$$

$$= 49.6\% \quad \underline{\underline{\text{Ans}}}$$

m-IV

$$M.P \times \frac{9}{10} \times \frac{4}{5} \times \frac{7}{10} = S.P$$

$$\frac{M.P}{125} = \frac{S.P}{63}$$

62

$$D\% = \frac{62}{125} \times 100$$

$$= 49.6\% \quad \underline{\underline{\text{Ans}}}$$

For: more than 3 Discount :-

Ex. 25%, 10%, 30%, $14\frac{2}{7}\%$

m-I	M.P	S.P
25% →	4	3
10% → $\frac{1}{10}$	10	9
30% = $\frac{3}{10}$	10	7
$14\frac{2}{7}\% = \frac{1}{7}$	7	6
	400	162

$$\frac{238}{400} \times 100 = \boxed{59.5\%}$$

m-II

$$M.P \times \frac{3}{4} \times \frac{9}{10} \times \frac{7}{10} \times \frac{6}{7} = S.P$$

$$\frac{M.P}{400} = \frac{S.P}{162}$$

$$\boxed{59.5\%} \text{ Ans}$$

- 16] A shopkeeper offers 10% discount on every purchase of an article. It also offers an additional discount of 12%, if the payment is made in cash. If the payment is made in original price of an item is ₹ 250, how much a customer will pay, if he wants to pay the price in cash?

Soln:- $250 \times \frac{9}{10} \times \frac{22}{25} = S.P$

$$S.P = \boxed{198} \text{ Ans}$$

- 17] A discount of 10% is offered on the price of an article if the payment is made online. An additional discount of 5% is given to credit card holders. A person wishes to buy a watch priced at Rs 60,000 by paying online through credit card. How much does he need to pay?

Soln $60,000 \times \frac{9}{10} \times \frac{19}{20} = S.P$

$$S.P = \boxed{54,300} \text{ Ans}$$

[18] An article is sold for ₹1,680 after two successive discounts of 20% and 16%. What is the marked price of the article?

Soln:-

$$M.P \times \frac{4}{5} \times \frac{21}{25} = 1680$$

$$M.P = \boxed{\text{₹}2500} \text{ Ans}$$

[19] The difference between successive discounts of 40% followed by 30% and 45% followed by 20% on the marked price of an article is ₹12. The marked Price of the article is ?

Soln:-

40% , 30%

45% , 20%

$$\Rightarrow 40 + 30 - \frac{40 \times 30}{100}$$

$$\Rightarrow 45 + 20 - \frac{45 \times 20}{100}$$

$$= 58\%$$

$$\Rightarrow 56\%$$

$$\text{Given} = \frac{\text{₹}12}{(58\% - 56\%)} \times 100 = \boxed{\text{₹}600} \text{ Ans}$$

[20] An article is sold for ₹288 after successive discounts of 25% and x%. If the marked Price of the article is ₹480, what is the value of x?

Soln:- [m-I]

$$480 \times \frac{3}{4} \times \frac{100-x}{100} = 288$$

$$100 - x = 80$$

$$x = \boxed{20\%}$$

Ans

[m-II]

$$\begin{array}{cc} M.P & S.P \\ 480 & : 288 \\ 25\% \rightarrow \frac{4}{4} & \times \frac{3}{3} \end{array}$$

$$480 \times 3 : 288 \times 4$$

$$5 : 4$$

$$\boxed{20\%} \text{ Ans}$$

[21] The marked price of a watch was ₹ 720. A man bought the same for ₹ 550.80 after getting two successive discounts. the first being 10%. then second discount rate is ?

Soln:- $720 \times \frac{9}{10} \times \frac{y}{x} = 550.80$

$$\frac{y}{x} = \frac{17}{20} \Rightarrow \text{D\%} = \frac{3}{20} \times 100 = \boxed{15\%} \text{ Ans}$$

[22] A person paid ₹ 17,000 for a motor-car after a single discount of 15%. If he is given single discount of 15%. If he is given successive discounts of 5% and 10% then how much he would pay?

Soln:- $M.p = \frac{17000}{85} \times 100 = 20,000$

after Discount = $20,000 \times \frac{19}{20} \times \frac{9}{10} \Rightarrow \boxed{₹ 17,100} \text{ Ans}$

[23] A dealer buys a table listed at ₹ 1,500 and gets successive discounts of 20% and 10%. He spends ₹ 20 on transportation and sells at a profit of 20%. Find the selling price of the table?

Soln:- $\left\{ \left[1500 \times \frac{4}{5} \times \frac{9}{10} \right] + 20 \right\} \times \frac{120}{100} = S.P$

$\underbrace{\hspace{10em}}_{S.P}$

$$(1080 + 20) \times \frac{6}{5} = S.P$$

$$S.P = \boxed{1320} \text{ Ans}$$

[24] A dealer buys an article listed at ₹100 and gets successive discounts of 10% and 20%. He spends 10% of the Cost Price on transportation. At what price should he sell the article to earn a profit of 15%?

Soln:- $\left\{ \left[100 \times \frac{9}{10} \times \frac{4}{5} \right] \times \frac{11}{10} \right\} \times \frac{23}{20} = \text{S.P}$

$$\text{S.P} = \boxed{\text{₹} 91.08} \text{ Ans}$$

[25] After giving two successive discount, each of $x\%$, on the marked price of an article, total discount is ₹259.20. If the marked price of the article is ₹720 then the value of x is ?

Soln (a) 18 (b) 24 (c) 20 (d) 25

$$\text{effective Discount} = \frac{259.20}{720} \times 100$$

$$x + x - \frac{x^2}{100} = 36\%$$

$$x = \boxed{20\%} \text{ Ans}$$

go through option

[26] The marked price of an article is ₹1500. If two successive discounts, each of $x\%$, on the marked Price is equal to a single discount of ₹587.40, then what will be the selling price of the article if a single discount of $x\%$ is given on the cost price?

Soln:- $\text{eff. discount} = \frac{587.40}{1500} \times 100$

$$x + x - \frac{x^2}{100} = 39.16\%$$

$$x = \boxed{22\%}$$

$$\begin{aligned} \text{C.p} &= 1500 \times \frac{78}{100} \\ &= \boxed{1170} \text{ Ans} \end{aligned}$$

Q7 The successive discounts of 20%, 10% and 8% is equivalent to a single discount of ?

Soln:-

M.P	S.P
5	4
10	9
25	23
1250	828

-422

$$\begin{aligned} 20\% &= \frac{1}{5} \\ 8\% &= \frac{2}{25} \\ 10\% &= \frac{1}{10} \end{aligned}$$

$$\text{Discount\%} = \frac{422}{1250} \times 100$$

$$= \boxed{33.76\%} \text{ Ans}$$

Q8 A double bed is marked at ₹7500. The shopkeeper allows successive discounts of 8%, 5% and 2% on it. What is the net selling price ?

Soln:-

$$S.P = 75000 \times \frac{23}{25} \times \frac{19}{20} \times \frac{49}{50}$$

$$8\% = \frac{2}{25}$$

$$5\% = \frac{1}{20}$$

$$S.P = \boxed{₹64239} \text{ Ans}$$

Q9 A bicycle, marked at ₹2,000 is sold with two successive discount of 20% and 10%. An additional discount of 5% is offered for cash payment. The selling Price of the bicycle at cash payment is ?

Soln:-

$$\text{selling price} = 2000 \times \frac{4}{5} \times \frac{9}{10} \times \frac{19}{20}$$

$$= \boxed{₹1368} \text{ Ans}$$

30 An article is marked at ₹ 5,000. The shopkeeper allows successive discount of $x\%$, $y\%$, $z\%$ on it. The net selling price is?

Soln:-

$$S.P = 5000 \times \frac{(100-x)}{100} \times \frac{(100-y)}{100} \times \frac{(100-z)}{100}$$

$$S.P = \frac{(100-x)(100-y)(100-z)}{100} \quad \underline{\text{Ans}}$$

31 A company offers three types of successive discounts:

(i) 25% and 15%.

(ii) 30% and 10%.

(iii) 35% and 5%

Which offer is the best for a customer?

Soln

$$\begin{aligned} \text{(i)} \quad 25 + 15 - \frac{25 \times 15}{100} &= 36.25\% \\ \text{(ii)} \quad 30 + 10 - \frac{30 \times 10}{100} &= 37\% \\ \text{(iii)} \quad 35 + 5 - \frac{35 \times 5}{100} &= 38.25\% \end{aligned}$$

Ans:- 3rd offer is the best for customer.

32 Which of the following successive discounts is better to a customer?

(i) 20%, 15%, 10%

(ii) 25%, 12%, 8%

Soln $\Rightarrow$

$$\text{(i)} \quad (20 + 15 + 10) - \frac{(200 + 150 + 200)}{100} + \frac{3000}{10000} = 38.8\%$$

$$\text{(ii)} \quad (25 + 12 + 8) - \frac{(300 + 96 + 200)}{100} + \frac{25 \times 12 \times 8}{10000} = 39.28\%$$

IInd case is better for the customer.

Type

33 A shopkeeper marks up his wares by 125% and offers 25% discount. What will be the selling if the cost price (in Rs.) is Rs. 640?

Soln:- m-I

$$C.P \times \frac{9}{4} \times \frac{3}{4} = S.P$$

$$640 \times \frac{9}{4} \times \frac{3}{4} = S.P$$

$$S.P = \boxed{1080} \text{ Ans}$$

m-II

C.P
4
4
16
↙ x40
640

125% = $\frac{5}{4}$ +

25% = $\frac{1}{4}$ -

S.P
9
3
27
↘ x40
<u>1080</u> Ans

m-III

$$\begin{aligned} \text{Overall Profit \%} &= A + B + \frac{AB}{100} \\ &= +125 - 25 + \frac{(125)(-25)}{100} \\ &= 100 - \frac{125}{100} = \frac{275}{4} \% \text{ Profit} \end{aligned}$$

$$C.P = 640$$

$$\text{Profit} = 640 \times \frac{275}{4 \times 100} = \underline{440}$$

$$S.P = C.P + \text{Profit}$$

$$= 640 + 440$$

$$= \boxed{1080} \text{ Ans}$$

m-IV Basic

+125% = 800

-25% = -360

C.P	M.P	S.P
640	1440	<u>1080</u> Ans

34 A trader marks the products 25% above the cost Price and allows a discount of 15%. If the cost Price is ₹ 2,080, then the selling Price is?

Soln:- **m-I**

$$2080 \times \frac{5}{4} \times \frac{17}{20} = S.P$$

$$S.P = \boxed{\text{Rs. 2210}}$$

Ans

m-II	C.P	S.P
	4	8
	+ 20	17
	16	17
↙ × 130	2080	↘ × 130
		2210
		<u>Ans</u>

35 A shopkeeper increases the price of an object by 40% and then sells it at 25% discount on the marked Price. If the selling price of such an object be ₹ 2100, its cost price for the shopkeeper was?

Soln:- **m-I**

$$x \times \frac{7}{5} \times \frac{3}{4} = 2100$$

$$x = \boxed{2000}$$

Ans

m-II	C.P	S.P
	5	7
	+ 40	3
	20	21
↙ × 100	2000	↘ × 100 → 2100
	2000	
	<u>Ans</u>	

m-III % Profit = $A + B + \frac{AB}{100}$

$$= +40 - 25 - \frac{25 \times 40}{100}$$

% Profit = 5%

$$C.P = \frac{2100}{105} \times 100 = \boxed{2000}$$

Ans

36 Selling Price of an article is ₹2,28,528. A shop-keeper marks up 15% more than cost Price and gives discount of 10%. Find out the cost Price?

Soln:-

$$C.P \times \frac{23}{20} \times \frac{9}{10} = ₹2,28,528$$

$$C.P = \boxed{220800} \text{ Ans}$$

37 A shopkeeper marks his goods at 40% more than their cost price and allows a discount of 25% on the marked Price. His gain or loss percent is?

Soln:- $\left[A + B + \frac{AB}{100} \right]$

$$+40 - 25 - \frac{40 \times 25}{100}$$

$$\boxed{+5\% \text{ Profit}} \text{ Ans}$$

m-II

C.P	S.P
5	7
<u>4</u>	<u>3</u>
20	21

$$\frac{1}{20} \times 100 = \boxed{5\% \text{ Profit}}$$

38 Sujata marks an article 36% above the cost Price and allows a 40% discount on the marked Price. The loss percentage is?

Soln:-

$$+36 - 40 - \frac{36 \times 40}{100}$$

$$\Rightarrow \boxed{-18.4\% \text{ Loss}} \text{ Ans}$$

m-II

C.P	S.P
25	34
<u>5</u>	<u>3</u>
125	102

$$\text{Loss \%} = \frac{-23}{125} \times 100$$

$$= \boxed{18.4\% \text{ Loss}} \text{ Ans}$$

- 39] A person marks his goods $x\%$ above the cost price and allows a discount of 30% on the marked price. If his profit is 5% , then the value of x will be?

Soln:- m-I $A + B + \frac{AB}{100} = \text{Profit/Loss}\%$

$$x - 30 - \frac{30x}{100} = 5$$

$$\frac{70x}{100} = 35$$

$$x = \boxed{50} \text{ Ans}$$

m-II

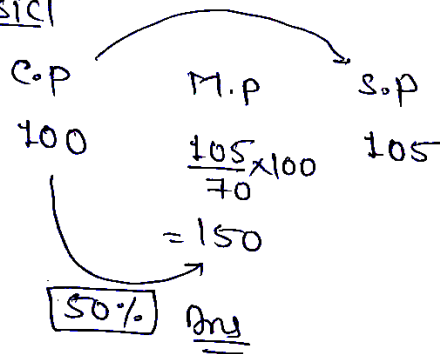
$$\begin{array}{l} \text{Profit}\% \rightarrow \frac{20}{10} : \frac{21}{7} \\ 30\% \downarrow \rightarrow \end{array}$$

$$\Rightarrow 2 : 3$$

$$\frac{1}{2} \times 100$$

$$= \boxed{50\%} \text{ Ans}$$

m-III Basic



- 40] The cost of manufacture of a tape recorder is Rs. 1,500. The manufacturer fixed the marked Price 20% above the cost of manufacture and allows a discount in such a way as to get a profit of 8% . The rate of discount is?

Soln:- $20 - x - \frac{20x}{100} = 8$

$$12 = \frac{120x}{100}$$

$$\text{Discount}\% = x = \boxed{10\%} \text{ Ans}$$

m-II

$$8\% = \frac{2}{25} \Rightarrow$$

$$\begin{array}{l} \text{Profit} \rightarrow \frac{5}{25} : \frac{9}{27} \\ 20\% \uparrow \rightarrow \end{array}$$

$$\Rightarrow 10 : 9$$

$$\text{Discount}\% \Rightarrow \frac{1}{10} \times 100$$

$$\Rightarrow \boxed{10\%} \text{ Ans}$$

Type

$$\frac{C.P}{(100 - D\%)} = \frac{M.P}{(100 \pm P/L\%)}$$

[41] The marked Price of an article is ₹ 800 and it is sold at a discount of 19%. If there is a gain of 8%, then by what percent above the cost price was the article marked?

Soln

$$\frac{C.P}{(100 - 19)} = \frac{M.P}{(100 + 8)}$$

$$\Rightarrow \frac{C.P}{81} = \frac{M.P}{108}$$

$$\frac{27}{81} \times 100 = \boxed{33\frac{1}{3}\%} \quad \underline{\underline{\text{Ans}}}$$

[42] A trader wishes to gain 20% after allowing 10% discount on the marked price to his customers. At what percent higher than the cost price must he mark his goods?

Soln:-

$$\frac{C.P}{90} = \frac{M.P}{120}$$

$$\frac{30}{90} \times 100 = \boxed{33\frac{1}{3}\%} \quad \underline{\underline{\text{Ans}}}$$

43 A trader sells his goods at a discount of 20%. He still makes a profit of 25%. If he sells the goods at the marked price only. His profit will be ?

Soln:-

$$\frac{C.P}{80} = \frac{M.P}{125}$$



$$\frac{45}{80} \times 100 = \boxed{56.25\%} \text{ Ans}$$

44 A photographer allows a discount of 10% on the advertised price of a camera. The price that must be marked on the camera, which cost him ₹600 to make a profit of 20% would be ?

Soln:-

$$\frac{C.P}{90} = \frac{M.P}{120}$$

600 ←
 Given

$$M.P = \frac{600}{90} \times 120$$

$$= \boxed{800} \text{ Ans}$$

45 A watch dealer pays 10% customs duty on a watch which costs Rs. 500 abroad. He desires to make a Profit of 20% after giving a discount of 25% to the buyer. The marked price should be ?

Soln:-

$$C.P = 500 + 500 \times \frac{10}{100}$$

$$C.P = 550$$

$$\frac{C.P}{75} = \frac{M.P}{125}$$

550 ←
 Given

$$M.P = \frac{550}{75} \times 125$$

$$= \boxed{880} \text{ Ans}$$

[46] An article of cost price ₹ 8,000 is marked at ₹ 11,200. After allowing a discount of x percent a profit of 12% is made. the value of x is ?

Soln:-

$$\frac{8000}{100-x} = \frac{11200}{100+12}$$

$$5 \times 112 = 7(100-x)$$

$$560 = 700 - 7x$$

$$x = \boxed{20\%} \text{ Ans}$$

[47] A shopkeeper marks an article at a price such that after giving a discount of 25%, the gains $x\%$. If the cost Price and the marked price of the article are ₹ 460 and ₹ 736 respectively, what is the value of x ?

Soln:-

$$\frac{460}{(100-25)} = \frac{736}{(100+x)}$$

$$5(100+x) = 8 \times 75$$

$$x = \boxed{20\%} \text{ Ans}$$

[48] When a shopkeeper gives 10% discount on the list price of a toy, his gain is 20%. If he had given a discount of 20% his percentage of gain would have been,

Soln:-

$$\frac{C.P}{90} = \frac{M.P}{120}$$

$$\downarrow -20\% = 24$$

$$96$$

$$\% \text{ gain} = \frac{6}{90} \times 100$$

$$= \boxed{6\frac{2}{3}\%} \text{ Ans}$$

[49] If a merchant offers a discount of 20% on the list Price, then she makes a loss of 16%. What % Profit or % loss will she make if she sells goods at a discount of 10% of the list price?

Soln:-

$$\frac{C.P}{80} = \frac{M.P}{84}$$

$\downarrow -10\% = -8.4$
 $\downarrow$
 75.6

$\swarrow -4.4$

$$\% \text{ Loss} = \frac{4.4}{80} \times 100 = \boxed{5.5\% \text{ Loss}}$$

[50] A dealer makes a profit of 20% even after giving a 10% discount on the advertised price of a scooter, if he makes a profit of Rs. 7500 on the sale of the scooter, the advertised price was?

Soln

$$C.P = \frac{7500}{20\%} \times 100\%$$

$$C.P = 37500$$

$$\frac{C.P}{90} = \frac{M.P}{120}$$

$$M.P = \frac{37500 \times 120}{90} = \boxed{50,000} \text{ Ans}$$

[51] A shopkeeper allows 23% Commission on his advertised Price and still makes a profit of 10%. If he gains Rs 56 on one item, his advertised price of the item, in Rs. is?

Soln

$$C.P = \frac{56}{10\%} \times 100\%$$

$$= \boxed{560}$$

$$\frac{C.P}{77} = \frac{M.P}{110}$$

$$M.P = \frac{560 \times 110}{77} = \boxed{800} \text{ Ans}$$

[S2] An article listed at ₹ 800 is sold at successive discounts of 25% and 15%. The buyer desires to sell it off at a profit of 20% after allowing a 10% discount. What would be his list price?

Soln:-

$$\text{Buyer C.P} = 800 \times \frac{3}{4} \times \frac{17}{20} = \boxed{510}^{\text{C.P}}$$

$$510 \leftarrow \frac{\text{C.P}}{90} = \frac{\text{M.P}}{120}$$

$$\text{M.p} = \frac{510}{90} \times 120 = \boxed{680} \text{ Ans}$$

[S3] The marked price of an article is Rs. 550. A shopkeeper allows a discount of 20% and still gets a Profit of 10%. If he sells it for Rs. 470, his profit Percent will be?

Soln:-

$$\frac{\text{C.P}}{80} = \frac{\text{M.P}}{110} \xrightarrow{\times 5} \frac{400}{\text{C.p}} \quad \text{Given } 550 \xrightarrow{\times 5}$$

C.P	S.P
400	470

$$\% \text{ Profit} = \frac{70}{400} \times 100 = \boxed{17.5\%} \text{ Ans}$$

[S4] A trader gains 25% by selling an article with 20% discount on its marked price. If the cost price of the article increased by 30%, then how much discount (in %) should he offer on the same marked price to gain 15% of profit?

Soln:-

$$\frac{\text{C.P}}{80} = \frac{\text{M.P}}{125} \quad \begin{array}{l} \text{+30\%} \downarrow = 24 \\ 104 \end{array} \xrightarrow{\begin{array}{l} \text{+15.6\%} \\ \text{+15\%} \end{array}} \frac{119.6}{\text{S.p}}$$

$$\% \text{ discount} = \frac{5.4}{125} \times 100 = \boxed{4.32\%} \text{ Ans}$$

Type

[55] If 3 dupatta is offered free on purchase of 6 dupattas. What is the effective discount on each dupattas.

Soln:-

Buy 6 get 3 free

$$\text{discount\%} = \frac{3}{9} \times 100$$

$$= \boxed{33\frac{1}{3}\%} \text{ Ans}$$

[56] Ticket for an adult is Rs. 1600 and a child is Rs. 600. 1 child goes free with two adults. If a group has 17 adults and 7 children what is the discount the group gets?

Soln:-

Free child = 7

$$\begin{aligned} \text{Then price of 7 children ticket} &= 7 \times 600 \\ &= 4200 \end{aligned}$$

And total price of 17 adult and 7 children

$$= 17 \times 1600 + 7 \times 600$$

$$= 31,400$$

$$\text{Then Required discount\%} = \frac{4200}{31400} \times 100$$

$$= \boxed{13.37\%} \text{ Ans}$$

[57] A shop-keeper sells a badminton racket whose marked price is ₹ 30, at a discount of 15% and gives a shuttle cock costing ₹ 1.50 free with each racket. Even then he makes a profit of 20%. His cost price, per racket is.

Soln:-

$$M.P = 30$$

$$S.P \text{ after discount} = 30 \times \frac{85}{100}$$

$$= 25.5$$

$$S.P \text{ of racket} = 25.5 - 1.5$$

$$= 24$$

$$C.P = \frac{24}{120} \times 100 = \boxed{20} \text{ Ans}$$

[58] A tradesman gives 4% discount on the marked price and gives 1 article free for buying every 15 articles and thus gains 35%. The marked price is increased above the cost price by?

Soln:-

$$\text{Let the M.P of 1 article} = ₹ 1$$

$$M.p \text{ of 15 article} = ₹ 15$$

$$S.p \text{ of } (15+1) \text{ article} = 15 \times \frac{96}{100} = ₹ 14.40$$

$$C.p \text{ of 16 article} = \frac{14.40}{135} \times 100$$

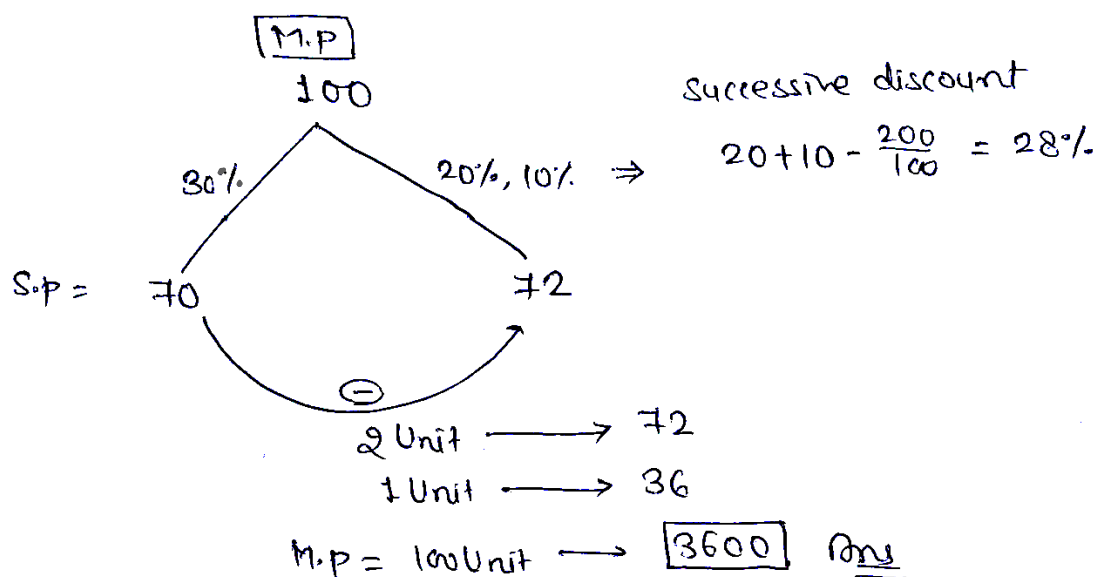
$$= \frac{32}{3}$$

$$C.P \text{ of 1 article} = ₹ \frac{2}{3}$$

$$\% = \frac{1 - \frac{2}{3}}{\frac{2}{3}} \times 100 = \boxed{50\%} \text{ Ans}$$

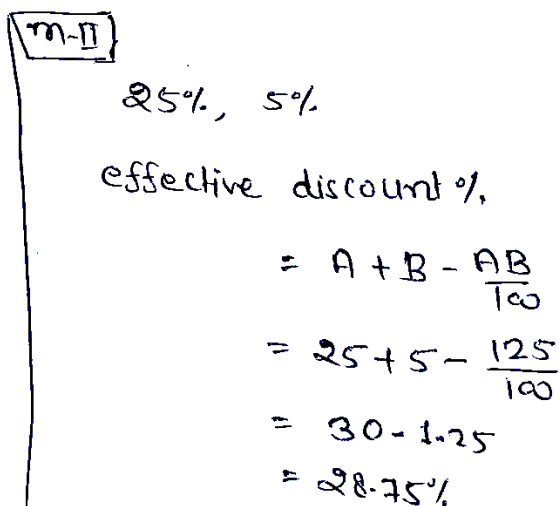
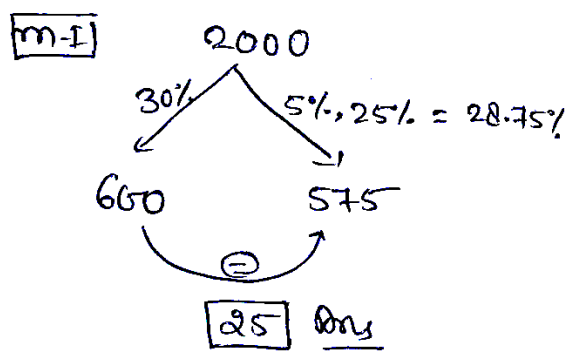
59] If on a marked price the difference of selling prices with a discount of 30% and two successive discount of 20% and 10% is 72, then the marked price is.

Soln



60] The difference between a discount of 30% on ₹2000, and two successive discounts of 25% and 5% on the same amount is ?

Soln:-



$$30\% - 28.75\% = 1.25\%$$

$$2000 \times \frac{1.25}{100} = [25] \text{ Ans}$$

[61] If a person purchases a shirt, he gets 6% discount. However, if he purchases two shirts, he gets 5% on the first and 8% on the second. If the price paid by person for two shirts is ₹925, then what will be the marked Price of each shirt?

Soln:-

$$M.P = 100x$$

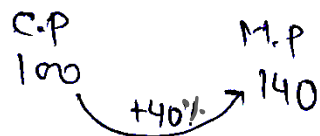
$$S.P \text{ of both} = 95x + 92x \\ = 187x \longrightarrow 925$$

$$x \longrightarrow \frac{925}{187} = 4.94$$

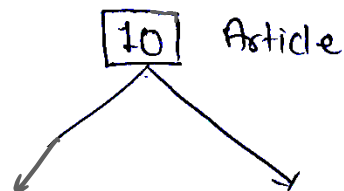
$$M.P = 100x = 4.94 \times 100 = \boxed{494} \text{ Ans}$$

[62] A trader marks his goods at 40% above the cost Price. He sells 70% of the cost price goods at the marked price and the rest. he sells by allowing a 40% discount on the marked price. His percentage profit is?

Soln:-



$$70\% \Rightarrow \frac{7}{10}$$



$$S.P = 7 \times 140 + 3 \times 140 \times \frac{60}{100} = 980 + 252 \\ = 1232$$

$$\text{Total Cost Price} = 1000 \quad \therefore S.P \quad 1232$$

$$\xrightarrow{23.2\%} \text{ Ans}$$

[63] Radha marks her goods 25% above the cost Price. She sells 35% of goods at the marked price, 40% at 15% discount and the remaining at 20% discount. What is her overall percentage gain? [SSC CGL Mains-2019]

Soln:-

C.P	M.P
400	500

$$25\% = \frac{1}{4}$$

$$\begin{aligned} \text{No. of goods} &\rightarrow 35 : 40 : 25 \\ &= 7 : 8 : 5 \end{aligned}$$

$$\begin{aligned} \text{Total S.p} &= 7 \times 500 + 8 \times 425 + 5 \times 400 \\ &= 3500 + 2000 + 2000 \end{aligned}$$

$$\text{Total S.p} = 8900$$

$$\text{Total C.p} = 400(7+8+5) = 8000$$

$$\% \text{ Profit} = \frac{8900 - 8000}{8000} \times 100 = \boxed{11.25\%} \text{ Ans}$$

[64] A shopkeeper listed the price of goods at 30% above the cost price. He sells half of the stock at listed price, one fourth of the stock at a discount of 15% and the remaining at 30% discount. His overall profit is.

Soln:-

C.P	M.P
100	130

Let article 4

$$\begin{array}{c} \swarrow \frac{1}{2} \quad \downarrow \frac{1}{4} \quad \searrow \frac{1}{4} \\ 2 \quad 1 \quad 1 \\ \downarrow \quad \downarrow -15\% \quad \downarrow -30\% \\ \text{Total S.p} = 260 + 110.5 + 91 \Rightarrow 461.5 \end{array}$$

$$\text{Total C.p} = 4 \times 100 = 400$$

$$\% \text{ Profit} = \frac{461.5 - 400}{400} \times 100 = \boxed{15\frac{3}{8}\%} \text{ Ans}$$

Q5 The discount offered on a shirt of Rs 500 and a pair of Trousers of Rs 1000 is 20% and 40% respectively. If Ajay bought 2 shirt and 3 pair of Trousers then what was the effective discount (in %) he recieved?

Soln:-

	shirt	Trousers
	500	1000
	↓ 20%	↓ 40%
Discount =	100	400

$$\text{Total Discount} = 2 \times 100 + 3 \times 400 = 200 + 1200 \\ = 1400$$

$$\text{Total M.p} = 2 \times 500 + 3 \times 1000 \\ = 4000$$

$$\% \text{ discount} = \frac{1400}{4000} \times 100 = \boxed{35\%}$$

Q6 A Rs. 720 tin of cheese is offered at 8% discount and a Rs. 1250 tin of butter at 20% discount. If we buy 5 tins of cheese and 3 tins of butter, what is the effective discount we get (in %) ?

Soln:-

720	1250
↓ 8%	↓ 20%
57.6	250

$$\text{Total Discount} = 5 \times 57.6 + 3 \times 250 \\ = 1038$$

$$\text{Total M.p} = 720 \times 5 + 1250 \times 3 \\ = 7350$$

$$\% \text{ Discount} = \frac{1038}{7350} \times 100 = \boxed{14.12\%} \quad \underline{\text{Ans}}$$

67

Type - Tax

A shopkeeper list the price of an article as ₹ 500. But he gives a certain discount which allows the buyer to pay ₹ 500 for the article including 10% sales tax. The rate of discount is ?

Soln:-

$$\begin{array}{ccc} \text{M.P} & & \text{S.P}_1 \xrightarrow{10\% \text{ Tax}} \text{S.P}_2 \\ 500 & : & \frac{500}{90} \times 100 \quad 500 \end{array}$$

$$9 : 10$$

$$\text{Discount} = \frac{1}{9} \times 100 = \boxed{11\frac{1}{9}\%} \text{ Ans}$$

68 The price of a product after getting 20% discount is Rs. 3,024 which includes 5% tax on selling price. What was the marked price (in Rs) of the product?

Soln:- m-I Basic

$$\begin{array}{ccc} \text{M.P} & \text{S.P}_1 & \text{S.P}_2 \\ \frac{3024 \times 20}{21} \times \frac{4}{5} & \xrightarrow{20\% \downarrow} & \frac{3024 \times 20}{21} \times 5\% \uparrow \rightarrow 3024 \\ = \boxed{3600} \text{ Ans} \end{array}$$

m-II

$$\text{M.P} = x$$

$$x \times \frac{4}{5} \times \frac{21}{20} = 3024$$

$$x = \boxed{3600} \text{ Ans}$$

m-III

$$\begin{aligned} &A + B + \frac{AB}{100} \\ &- 20 + 5 - \frac{20 \times 5}{100} \\ &- 16\% \end{aligned}$$

$$\text{M.P} \times \frac{84}{100} = 3024$$

$$\text{M.P} = \boxed{3600} \text{ Ans}$$

69 A Publisher printed 2000 copies of a book at a cost of Rs. 70,000. He distributes 400 copies free as specimen copies. He gave 30% discount on marked price of each book which is Rs. 75. What is his gain or loss percentage?

Soln:- CP of 1 book is $= \frac{70,000}{2000} = \text{Rs. } 35$

M.P of 1 book $= ₹ 75$

S.p after 30% Discount $= \text{Rs. } 52.5$

S.p of 1600 books $= 52.5 \times 1600$
 $= \text{Rs. } 84000$

% Profit $= \frac{84000 - 70000}{70,000} \times 100$
 $= \boxed{20\% \text{ gain}} \quad \underline{\underline{\text{Ans}}}$

70 A trader gives a discount of 4% for purchases above ₹ 25,000, 6% for purchases above ₹ 35,000 and 8% for purchases above ₹ 50,000. If an item is purchased for ₹ 38,500. what would be the amount of discount?

Soln:- If an item purchased $= \text{Rs } 38,500$

so. Discount % $= 6\%$

Amount of discount $= 38,500 \times \frac{6}{100}$

$= \boxed{\text{Rs. } 2310} \quad \underline{\underline{\text{Ans}}}$